Psychiatry Research 360 (2026) 117085



Contents lists available at ScienceDirect

Psychiatry Research

journal homepage: www.elsevier.com/locate/psychres



Review article



# Prevalence and moderators of personality disorders in adults with ADHD: A meta-analysis

Dimitrios Adamis [a,*], Tianran Zhang [b], Blánaid Gavin [c], Fiona McNicholas [d]

[a] Consultant Psychiatrist, Sligo Mental Health Services Clarion Rd Sligo, Ireland, University of Galway, University College Dublin and University of Limerick, Ireland
[b] Psychologist, PhD candidate, Atlantic Technological University, Sligo, Clarion Rd Sligo, Ireland
[c] University College Dublin, Ireland
[d] University College Dublin, Lucena CAMHS Rathgar, Dublin 7, CHI Crumlin Dublin 12. Ireland

ARTICLE INFO

*Keywords:*
ADHD
Personality disorders
Psychiatric comorbidity
Meta-analysis
Diagnostic assessment
SCID-II
MCMI-III

ABSTRACT

This meta-analysis investigated the prevalence and moderators of personality disorders (PDs) in adults with Attention-Deficit/Hyperactivity Disorder (ADHD), synthesizing data from 11 studies encompassing 2120 participants across clinical and nonclinical settings. The results indicate a substantial but highly variable burden of personality pathology, with a pooled prevalence for the presence of at least one comorbid PD estimated at 57 % (95 % CI [42–71]). However, this figure was largely driven by specialized outpatient populations, and statistical heterogeneity was exceptionally high ($I^2 > 90$ %), suggesting that prevalence is deeply context-dependent. Among specific diagnoses, the highest co-occurrence rates were observed for Passive-Aggressive (25.3 %), Avoidant (23.1 %), and Borderline (21.9 %) personality disorders. Moderator analyses revealed that these estimates were significantly influenced by diagnostic methodology; structured clinical interviews (such as the SCID-II) and self-report inventories (like the MCMI) consistently yielded higher rates than standard clinical assessments. Furthermore, setting played a critical role, with ADHD-specific clinics showing significantly higher coexistence than prison or student samples. While tests for publication bias were non-significant, they were limited by low statistical power. Ultimately, these findings underscore the high frequency of personality-related impairment in adult ADHD while highlighting the challenges of "diagnostic noise" and phenotypic overlap. The extreme fluctuations in reported rates suggest that clinicians must look beyond categorical labels, acknowledging that the identified prevalence is often a byproduct of the specific instruments and clinical frameworks employed.

## 1. Introduction

Attention Deficit-Hyperactivity Disorder (ADHD) is a neurodevelopmental disorder with onset in childhood, characterized by symptoms of inattention, hyperactivity, impulsivity, or a combination of these. Evidence indicates that ADHD frequently persists into adulthood, with approximately 60 % of cases remaining impairing (Kooij et al., 2010; Turgay et al., 2012). The clinical presentation, however, changes across the lifespan: hyperactivity and impulsivity symptoms often decline with age, while inattentiveness tends to persist, alongside common co-occurring problems such as mood dysregulation, other mental disorders, and personality disorders (PDs) (Biederman et al., 2020; Cherkasova et al., 2022; Yadav et al., 2021). Personality Disorders are commonly reported in adults with ADHD. PDs are characterized by enduring inflexible and maladaptive patterns of behaviour, cognition, and inner experience, evident across many contexts and culturally atypical. These enduring patterns typically emerge early in development and are associated with significant distress or functional impairment (APA 2013).

Research examining the prevalence of PDs in adults with ADHD—including systematic reviews and meta-analytic work—has employed heterogeneous methodological approaches and theoretical frameworks. Much of the existing literature has relied on dimensional personality models, most notably Cloninger's biopsychosocial model and the Five-Factor Model (Gomez and Corr, 2014), thereby emphasising trait-level personality profiles rather than formal diagnostic categorisation. While such dimensional approaches offer valuable insight into the structure and variability of personality characteristics in ADHD,

* Corresponding author.
*E-mail addresses:* dimaadamis@yahoo.com (D. Adamis), tianran.zhang@research.atu.ie (T. Zhang), blanaid.gavin@ucd.ie (B. Gavin), Fionamcn2008@gmail.com (F. McNicholas).

https://doi.org/10.1016/j.psychres.2026.117085
Received 14 December 2025; Received in revised form 2 March 2026; Accepted 9 March 2026
Available online 10 March 2026
0165-1781/© 2026 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

they do not directly quantify the burden of categorically defined PDs. For instance, Jacobsson et al. (2021) reported substantial heterogeneity (85 %–97 %) across Neuroticism, Extraversion, Openness, Agreeableness, and Conscientiousness, featuring considerable within-group variability in personality traits among adults with ADHD. However, because dimensional models assess personality traits along normative continua rather than diagnostic thresholds, such findings cannot be directly extrapolated to estimate PD prevalence. By contrast, both the DSM and ICD categorise PDs as maladaptive, persistent, and functionally impairing patterns, and DSM-IV criteria have demonstrated stronger associations with functional impairment compared to dimensional trait scores (Skodol et al., 2005). Despite the clinical relevance of categorical diagnoses, available research remains limited and methodologically inconsistent, with reported PD prevalence among adults with ADHD ranging from 10 % to 75 % (Matthies and Philipsen, 2016).

In addition to conceptual differences between categorical and dimensional models, the reliability and validity of PD diagnosis depend critically on the assessment method employed. Unstructured clinical diagnoses have been shown to demonstrate limited interrater reliability and are prone to both under- and over-diagnosis compared with structured or semi-structured interviews (Pilkonis et al., 1991; Zimmerman, 1994). Semi-structured interviews such as the Structured Clinical Interview for DSM-IV Axis II Personality Disorders (SCID-II) are generally considered the gold standard for categorical PD assessment and demonstrate acceptable to good interrater reliability and construct validity (Lobbestael et al., 2011; Maffei et al., 1997). Consequently, variability in diagnostic procedures across studies is likely to substantially influence reported prevalence estimates. Careful consideration of diagnostic methodology is therefore essential when synthesising PD prevalence in adults with ADHD.

Taken together, these conceptual and methodological differences highlight the need for systematic studies using standardised, diagnostically anchored assessments to clarify the prevalence and clinical relevance of PDs in adults with ADHD. Therefore, given the lack of robust prevalence estimates the primary aim of this review was to estimate the prevalence of PDs among adults with ADHD using categorical diagnostic definitions. Secondary aims were to examine whether study setting, diagnostic instrument, and age moderated prevalence estimates.

## 2. Methods

### 2.1. Search strategy

The review followed Preferred Reporting Items for Systematic Reviews and Meta-Analyses statement (PRISMA) guidelines (Moher et al., 2009). The databases Cochrane Library, PubMed, PsycINFO, EBSCO, CINAHL Complete, Scopus and EMBASE were searched by using Boolean operators of the following keywords: (ADHD OR ADD OR ADDH OR Attention Deficit Disorder with Hyperactivity OR Attention Deficit Hyperactivity Disorder) AND (personality traits OR personality dimensions OR Personality disorders OR Temperament OR Millon Clinical Multiaxial Inventory). Although the primary aim of the review was to synthesise prevalence estimates based on categorical personality disorder (PD) diagnoses, broader dimensional terms were included to maximise search sensitivity. Several widely used instruments (e.g., Millon Clinical Multiaxial Inventory [MCMI]) generate dimensional scores but permit classification according to DSM-based diagnostic thresholds. However, only studies reporting categorical PD diagnoses, or providing sufficient data to derive them, were eligible for inclusion. Detailed research strategy in appendix 1. To ensure comprehensive coverage and mitigate the limitations of electronic database indexing, a supplementary manual search was conducted. This included citation tracking (backward and forward snowballing) of included papers and a review of literature known to the authors; any studies identified through this process were then screened against the same rigorous inclusion and exclusion criteria as the primary search results. Search results were exported to a spreadsheet (.csv) and to EndNote software for duplicate removal and screening.

### 2.2. Outcomes-types of data

The primary outcome was the point prevalence of any PD and of specific PDs, defined according to DSM criteria. When ICD criteria were used the personality diagnosis was converted to corresponding DSM.

### 2.3. Studies selection

Two reviewers independently performed the search, screen the title and abstract of all articles yielded by the keywords search. Records were excluded (grouped as irrelevant) at this stage if they did not: a) examine adult ADHD populations, b) if assess personality disorders or diagnosable personality pathology, or if report neurobiological studies without PD assessment. Review of the full-text articles of the selected studies were also evaluated independently. Any disagreement between them was resolved by consensus.

### 2.4. Inclusion criteria

Studies were included if they met all of the following conditions:

- Prospective observational studies (defined here as studies involving the active recruitment and clinical assessment of participants) and high-quality cross-sectional research. This approach ensured that prevalence estimates were derived from contemporary clinical cohorts rather than retrospective administrative database reviews, which often lack standardized diagnostic validation.
- Adult populations diagnosed with ADHD, as defined by the study.
- Clear definition of ADHD, using a *validated diagnostic scale* or *research diagnostic criteria*. The diagnostic integrity of the ADHD cohorts was maintained by excluding studies relying solely on retrospective chart reviews or unverified administrative data. Instead, inclusion was restricted to studies utilizing validated diagnostic instruments and expert clinical evaluation, typically within specialized psychiatric settings. Given this consistently high threshold for ADHD diagnosis across the included literature, diagnostic rigor was not treated as a statistical moderator, as the sample represented a relatively homogenous group of clinically confirmed cases.
- Use of validated diagnostic instruments or criteria for Personality Disorders, or provision of sufficient information to allow conversion of PD classifications to DSM-5 categories. Where formal DSM-5 labels were not provided, 'sufficient information' was defined as the reporting of standardized instrument scores (e.g., MCMI-III subscale scores) or ICD-10 codes that have a direct, documented algorithmic conversion to DSM-5 diagnostic categories.

Generally, studies were included if they utilized established diagnostic frameworks. For ADHD, this required a diagnosis via a semi-structured diagnostic interview (e.g., DIVA) or a validated clinical rating scale (e.g., CAARS, Conners) with scores exceeding established clinical cut-offs. For Personality Disorders, inclusion was restricted to studies using standardized diagnostic instruments (e.g., SCID, MCMI) or those providing categorical clinical diagnoses that allowed for a direct mapping onto DSM-5 categorical criteria by two independent reviewers.

- Availability of adequate statistical data to contribute to the meta-analysis.
- Published in English.

### 2.5. Exclusion criteria

Studies were excluded if they met any of the following conditions:

- Retrospective designs, case studies, case series, narrative reviews, systematic reviews, or commentaries.
- Samples restricted to a specific psychiatric disorder or specific PD subtype, such as studies conducted exclusively in populations with eating disorders, borderline personality disorder, or antisocial personality disorder.
- Conference abstracts or proceedings without a corresponding full publication.
- Duplicate publications, in which case the version with the larger sample size was retained.
- For longitudinal or cohort studies, only data corresponding to the adult time point with the largest sample size were included.

### 2.6. Quality appraisal and risk of bias

Selected studies were evaluated using the Joanna Briggs Institute (JBI) Critical Appraisal Checklist for Studies Reporting Prevalence Data (Munn et al., 2014) by two independent raters. The checklist is designated for prevalence studies and includes 9 items rated as yes, no, unclear and not applicable. The areas (items) that the checklist investigates are sample frame, sampling method, sample size, adequate description of subject and setting, data analysis, validity of diagnostic method, reliability, appropriate statistical analysis, and response rate. Any inconsistency between the two raters was resolved by consensus.

### 2.7. Data extraction and statistical analysis

From each included study the following data were extracted: first author and year of publication, country in which the research was conducted (nominal); design of the study (nominal); setting e.g., prison, outpatients, college etc. (nominal); final sample size analysed (numeric); age, (mean, standard deviation) (numeric); gender (nominal); scales /criteria for diagnosis of ADHD (nominal); number of participants with each subtype of ADHD (hyperactive, inattention, combined, numeric); criteria/scales for diagnosis of PDs (nominal), number of participants with PDs (numeric), number of participants without PDs (numeric) number of participants with specific PD (numeric). For studies in which some of the above data were not available and could not be extracted the corresponding author was contacted by email with a request to supply them.

Analysis was carried out with the 'metafor' and "meta" package of R software (Balduzzi et al., 2019; Viechtbauer, 2010) For the meta-analysis, a random-effects model was incorporated, because the studies include different populations. When moderators were included, a mixed-effects model was used. If a study reported median and interquartile range or range for one variable, the mean and standard deviation were calculated based on already published methods (Hozo et al., 2005; Wan et al., 2014). Proportions were transformed using the Freeman–Tukey double-arcsine method to stabilize variances prior to pooling, and the Der Simonian–Laird estimator was used for random-effects meta-analysis. In addition, homogeneity was investigated using the Q test and for the quantification of heterogeneity, the Higgins $I^2$. Bias in study availability (publication bias) was explored by visual inspection of funnel plots and using Egger's test. Influence diagnostics were used to examine the influence of each study on the results. For the meta-regression models (moderators) the Knapp-Hartung adjustment was used to obtain more reliable estimates (Higgins et al., 2002).

## 3. Results

Flow diagram of the selection of studies in Fig. 1

### 3.1. Description of the studies

Eleven studies ($k = 11$) fulfilled the inclusion/ exclusion criteria and included in this analysis. However, not all studies reported the total number of participants with any PD; therefore, the pooled prevalence estimate was restricted to five studies. Similarly, only one study reported separated PDs for gender and only 3 studies reported prevalence of PD across the three ADHD subtypes. Therefore, subgroup analyses for gender and subtypes were abandoned as it was not possible to extract data or calculate the prevalence of PD in each group. All the selected studies were observational and cross-sectional except one (Gift et al. 2016) which was longitudinal and data were extracted from the first

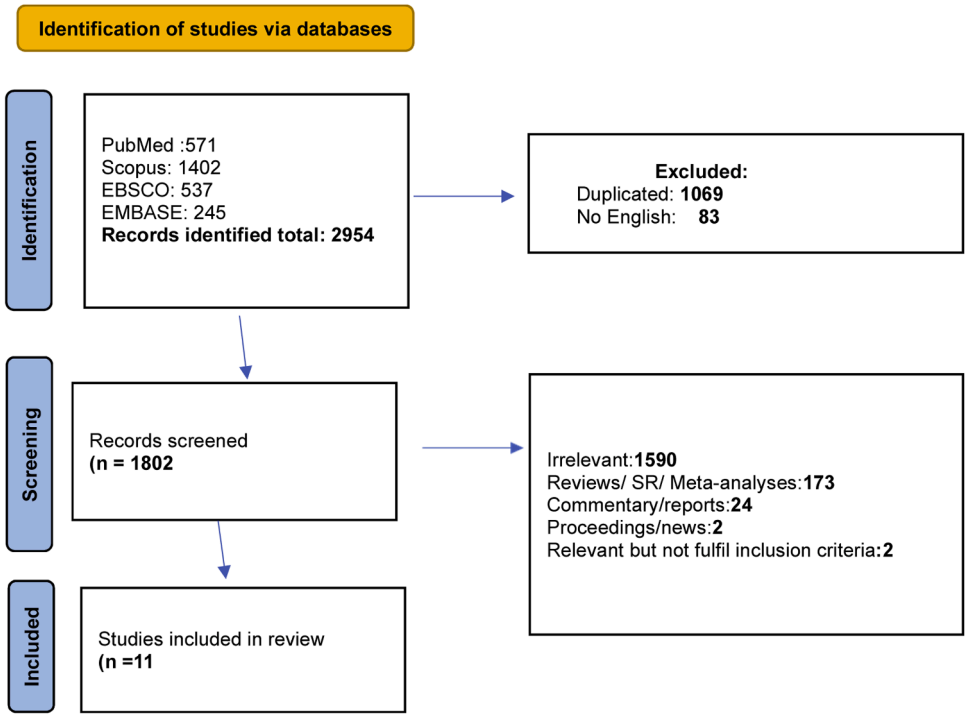



**Fig. 1.** Flowchart of the study.

assessment. Study characteristics are given in Table 1 and quality appraisal of included studies in the supplementary material (table S1). The total number of participants was 2120 of whom 855 were females (40.33 %). One study (Hamzeloo et al., 2016) was conducted in a male prison ($n = 147$), one ($n = 48$) in university students (Kavakçi et al., 2012), and another one (Young et al., 2003) did not reported gender distribution ($n = 30$).

### 3.2. Meta-analysis of studies

#### 3.2.1. Estimation of any PD prevalence

Five studies ($k = 5$, $N = 8$, $e = 461$) reported, or it was able the total number of PDs to be calculated. A total of 461 participants were identified to have at least one PD (event=$e$) out of 835 included in those five studies. Using a random-effects model, it shows that the pooled prevalence of any PD was 57 %, [95 % CI (42–71)]. The forest plot for the meta-analysis is shown in Fig. 2. The Q test for heterogeneity showed significant heterogeneity among the studies $Q = 72.28$, df: 4, $p < 0.001$ and the $I^2$= 94.5 %. The funnel plot shown no asymmetry, indicating no evidence publication bias. (Fig. 3). Egger's test was not significant ($z = 0.114$, $p = 0.909$), suggesting no evidence of publication bias. No study emerged as a clearly influential outlier across all metrics with the exception of Adamis et al. (2023) study which showed a studentized residual >2, suggesting potential deviation from the model fit. The plot of the influence diagnostics is presented in Figure S1.

*Moderators / meta-regression analysis:*

**Combined Model (Age, diagnostic PD scale, and setting)**: A mixed-effects meta-regression including all moderators (age, scales for diagnosis of PD and setting) could not be fitted due to model overfitting (number of parameters larger the number of studies). Therefore, models were fitted separately.

**Mean age as moderator:** A mixed-effects meta-regression was conducted using the restricted maximum likelihood (REML) estimator and the Knapp–Hartung adjustment to assess if age moderated the association between PDs and ADHD. Mean age was not a significant moderator of the prevalence of PDs adults with ADHD ($\beta = -0.03$, $p = 0.730$). The model showed high residual heterogeneity ($I^2 = 96.45$ %), indicating that age did not explain the substantial between-study variability.

**Scale for diagnosis of PD as moderator:** Similarly, a mixed-effects meta-regression was conducted to assess the moderating effects of scales used to diagnose PDs. The meta-regression indicated significant between-group differences in estimated prevalence rates ($F(3,2) = 57.15$, $p = 0.017$). The MCMI-III yielded the highest pooled prevalence ($\beta = 1.01$, $p = 0.039$), followed by SCID-II ($\beta = 0.77$, $p = 0.017$) and clinical assessments ($\beta = 0.73$, $p = 0.039$). This suggests that self-report and dimensional measures (e.g., MCMI-III) identify a broader range of PD features than clinician-administered assessments, which tend to be more conservative.

**Setting:** Because all the 5 studies were conducted in special ADHD clinics meta regression with setting as variable do not have any meaning.

#### 3.2.2. Estimation of schizotypal personality disorder prevalence

Eight studies have reported on Schizotypal PD ($k = 8$, $N = 1608$, $e = 83$) The pooled prevalence of schizotypal PD was 7.7 % (95 % CI: 1.1–18.8 %). (Figure S2). Heterogeneity was substantial ($Q = 236.08$, df:7, $p < 0.0001$; $I^2 = 97.0$ %), indicating considerable variability in prevalence estimates across studies. Funnel plot shown no asymmetry (Figure S3) and Egger's test was not significant ($z = 1.289$, $p = 0.197$). Both of them suggesting no publication bias. Influence diagnostics indicated that most studies contributed consistently to the pooled estimate, with only one study (Gift et al., 2016) showing a notable influence on the overall model. Excluding this outlier heterogeneity was substantially reduced ($\tau^2 = 0.021$). See also Figure S3.

*Moderators:* The combined model of the 3 variables together as moderators (age, scales, and setting) showed not any significant effect. ($F_{(5,3)} = 1.220$, $p = 0.464$) and increased the heterogeneity to 98.76 %. Similarly, age as moderator was not significant [$\beta$ estimate=0.0073, $t = 0.317$, df:6, $p = 0.762$, 95 % CI (−0.0492 to 0.0638)]. Scales also they did not have significant effect ($F(4,4) = 1.93$, $p = 0.270$) and either setting ($F(3,5) = 3.09$, $p = 0.130$).

#### 3.2.3. Estimation of Paranoid Personality Disorder prevalence

($k = 10$, $N = 1785$, $e = 220$). The pooled prevalence of paranoid PD among adults with ADHD was 11.9 % (95 % CI: 5.6–20.1), with substantial heterogeneity ($Q = 107.24$, df:9, $p < 0.001$; $I^2 = 94.3$ %), (forest plot in figure S6). Funnel plot (figure S6) and Egger's regression test ($z = 0.48$, $p = 0.63$) indicated no evidence of publication bias. Influence

**Table 1**
Studies' characteristics.

| Authors | Country | Design | Sample | N | M. A | SD | Scale PD | D PD | Scale ADHD | D ADHD | M | F |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Adamis et al. (2023) | Ireland | cross sectional | ADHD service | 147 | 32.97 | 11.56 | MCMI-III | DSM-IV | CAADID/clinical | DSM_5 | 73 | 74 |
| Cumyn et al. (2009) | Canada | cross sectional | ADHD service | 335 | | | SCID-II | DSM-IV | CAADID/clinical | DSM-IV | 209 | 126 |
| Edvinsson et al. (2013) | Sweden | cross sectional | ADHD service | 168 | 34.4 | 9.6 | SCID-II | DSM-IV | clinical | DSM-IV | 90 | 78 |
| Gift et al. (2016) | USA | longitudinal | ADHD service | 115 | 33.6 | 11.61 | Clinical | DSM-IV | WRAADDS/CAADID/ clinical | DSM-IV | 78 | 37 |
| Hamzeloo et al. (2016) | Iran | cross sectional | Prison | 147 | 29.8 | 6.1 | MCMI-III | DSM-IV | ASRS/clinical | DSM-IV | 147 | 0 |
| Irastorza Eguskiza et al. (2018) | Spain | cross sectional | General adult OPC | 78 | 34.6 | 10.5 | SCID-II | DSM-IV | CAADID | DSM-IV | 45 | 33 |
| Jacob et al. (2016) | Germany | cross sectional | General adult OPC | 878 | 34.5 | 10.2 | SCID-II | DSM-IV | DSM-IV symptom list | DSM-IV | 438 | 440 |
| Kavakçi et al. (2012) | Turkey | cross sectional | students | 48 | 21.4 | 2.3 | SCID-II | DSM-IV | Adult ADD ADHD/ MINI | DSM-IV | 28 | 20 |
| May and Bos (2000) | USA | cross sectional | ADHD service | 104 | 33 | | MCMI-II | DSM-III-R | DSM-III-R checklist | DSM-III-R | 78 | 26 |
| Oliva et al. (2020) | Italy | cross sectional | ADHD service | 70 | 30.8 | 11.53 | MCMI-III | DSM-IV | DIVA 2.0/clinical | DSM-IV | 49 | 21 |
| Young et al. (2003) | UK | cross sectional | ADHD service | 30 | 24 | | SAP | ICD-10 | DSM-IV/ clinical | DSM-IV | | |

OPC: outpatient clinic, M. A: Mean Age, SD: Standard Deviation, D PD: Diagnostic criteria for Personality Disorder D ADHD: diagnostic criteria for ADHD, M: male with ADHD, F: Females with ADHD, MCMI: Millon Clinical Multiaxial Inventory. SCID: Structured Clinical Interview for DSM Disorders, SAP: Standard Assessment of Personality, CAADID: Conners' Adult ADHD Diagnostic Interview for DSM-IV, WRAADDS: Wender-Reimherr Adult Attention Deficit Disorder Scale, DIVA: Diagnostic Interview for ADHD in adults, ASRS: Adult ADHD Self-Report Scale, MINI: Mini International Neuropsychiatric Interview.

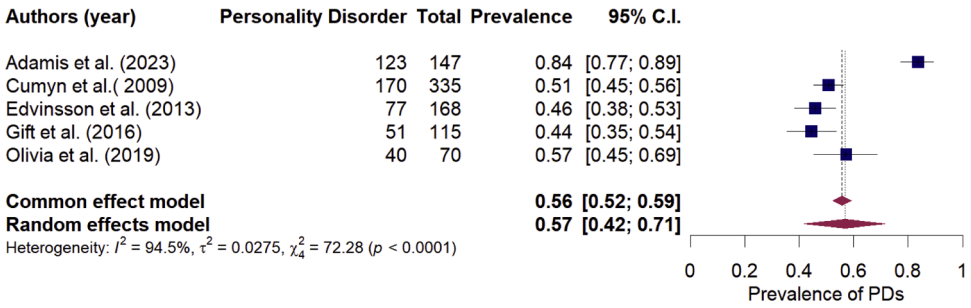



**Fig. 2.** Forest plot for studies examining prevalence of PDs among adult ADHD population.

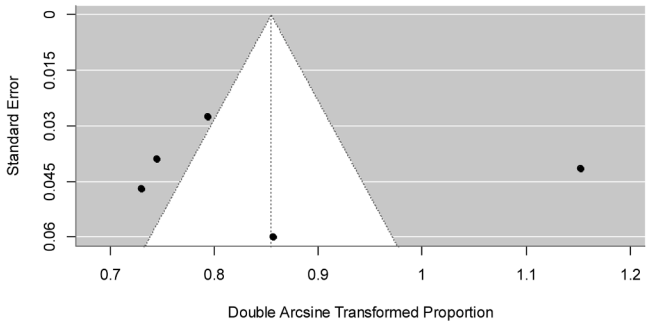



**Fig. 3.** Funnel plot assessing publication bias for studies investigating prevalence of PDs in adult ADHD population.

diagnostics (figure S7) suggested one study as potentially influential, (Hamzeloo et al., 2016). This study completed in prison with only males and when it excluded, this markedly reduced heterogeneity ($\tau^2$ = 0.0057).

### 3.2.4. Moderators

The combined model did not show any significant effect ($F_{(9,1)}$ = 1.611, $p$ = 0.549) and heterogeneity remained extremely high ($I^2$ = 98.28 %). Age also did not have significant effects in the model either [β estimate=0.011, $t$ = 0.812, df:8, $p$ = 0. 441, 95 % CI (−0.0201 −0.0418)] Among the diagnostic tools, only the SCID-II yielded a statistically significant association with higher observed prevalence rates (β = 0.41, $p$ = 0.016, 95 % CI [0.12, 0.71]). Studies based on ADHD clinical services (β = 0.33, $p$ = 0.002, 95 % CI [0.18–0.48]) and general adult outpatient samples (β = 0.58, $p$ = 0.0015, 95 % CI [0.32–0.83]) demonstrated significantly higher prevalence rates of paranoid PD compared to studies conducted in prison or student populations. It should be noted, however, that these results are subject to limited generalizability, as the prison and student categories were each represented by only a single study. Consequently, the observed differences in prevalence rates across these settings should be interpreted with caution due to the lack of comparative data within those specific subgroups. Estimation of Schizoid Personality Disorder prevalence

Based on nine studies ($N$ = 1638, $e$ = 85), the pooled random-effects prevalence of Schizoid PD was 6.4 % (95 % CI: 1.4 %–14.1 %). (figure S8). Heterogeneity was high ($Q$ = 157.67, df:8, $p$ < 0.001; $I^2$ = 95.1 %). No significant publication bias or small-study effects were detected ($z$ = 0.04, $p$ = 0.965 and funnel plot in figure S9). Influence diagnostics (figure S10) did not identify any study as potentially influential in the results.

*Moderators*: Combined model, and age alone did not have any significant effect in the prevalence while the MCMI-III scale yielded significantly higher prevalence estimates for Schizoid PD ($p$ = 0.043). Regarding the setting, Schizoid PD prevalence was higher in ADHD service samples ($p$ = 0.016) and marginally elevated in general adult outpatient samples ($p$ = 0.083) compared to student sample.

### 3.2.5. Estimation of borderline personality disorder prevalence

All the included studies reported data on Borderline PD ($k$ = 11, $N$ = 2120, $e$ = 421). The pooled prevalence of PD was 21.9 % (95 % CI: 12.4–33.1). The heterogeneity test was significant ($Q$ = 286.46, df:10, $p$ < 0.0001; $I^2$ = 96.5 %), figure S11. Funnel plot, (figure S12), and Egger's test ($z$ = 0.41, $p$ = 0.68), indicated no bias. Influence diagnostics (figure S13) identified one study as potentially influential, (Irastorza Eguskiza et al., 2018) which, when excluded, markedly reduced heterogeneity ($\tau^2$ = 0.014).

*Moderators:* Combined variables and age alone did not show any significant effect. Among the instruments, only the SCID-II yielded a statistically significant positive association with effect size (β = 0.58, $p$ = 0.011), whereas setting significantly moderated the association ($F_{(4,7)}$ = 9.62, $p$ = 0.006). Studies conducted in ADHD service and general adult outpatient clinics reported higher prevalence estimates (β = 0.40, $p$ = 0.007; β = 0.86, $p$ = 0.003, respectively) compared to those using prison or university settings.

### 3.2.6. Estimation of antisocial personality disorder prevalence

Eleven studies ($k$ = 11, $N$ = 2120, $e$ = 289) yielded a pooled prevalence estimate of 18.2 % (95 % CI: 10.3–27.5), figure S14. Heterogeneity was high ($I^2$ = 95.4 %, $Q$ = 224.69, df:10 $p$ < 0.001). The regression test for funnel plot (figure S15) asymmetry indicated no evidence of publication bias ($z$ = 1.09, $p$ = 0.280). Influence diagnostics (figure S16) indicated that no study influences the overall model estimates.

*Moderators:* Moderator analysis revealed that all the structured diagnostic scales demonstrated significantly higher prevalence estimates compared to clinical assessment (1 study), while the prevalence varied significantly across settings, with higher rates observed among prison samples ($p$ = 0.0042), ADHD service users ($p$ = 0.0004), and general adult outpatient samples ($p$ = 0.0106), whereas student samples did not differ significantly ($p$ = 0.190). Despite accounting for scales and setting heterogeneity persisted ($I^2$ = 93.89 %), indicating the presence of additional unexplained between-study variability. Combined variables and age alone did not have any effect on prevalence.

### 3.2.7. Estimation of histrionic personality disorder prevalence

The pooled prevalence of histrionic PD among individuals with ADHD ($k$ = 9; $n$ = 1638; $e$ = 271) was estimated at 7.3 % (95 % CI [1.0–17.9]) see figure S17. Heterogeneity was high ($Q$ = 258.91, df:8, $p$ < 0.0001; $I^2$ = 96.9 %), suggesting considerable variability across studies. The Egger's test for funnel plot asymmetry was nonsignificant ($z$ = −0.997, $p$ = 0.319), indicating no evidence of publication bias. See also figure S18. Influence diagnostics (figure S19) indicated that one study (Jacob et al., 2016) exerted an unequal influence on the overall model fit.

*Moderators:* Combined Model and age alone did not show significant effects. Scales shown significant overall moderator effect ($F_{(5,4)}$ = 10.94, $p$ = 0.0189), and MCMI-II ($p$ = 0.0045) and SCID-II ($p$ = 0.0132) showed significant positive associations with the effect size. Histrionic PD prevalence estimates may vary across settings, with clinical samples

tending to yield higher rates but in marginal significancy. In all the moderators analysis the heterogeneity remained high $I^2 > 91.65$ %.

### 3.2.8. Estimation of narcissistic personality disorder prevalence

The random-effects meta-analysis, ($k = 9$; $n = 1943$; $e = 368$), estimated a pooled prevalence of 13.5 % (95 % CI [2.5- 30.7]) for the narcissistic PD, (figure S20). Substantial heterogeneity was observed ($Q = 383.87$, df:8, $p < 0.0001$; $I^2 = 98.64$ %), and no evidence of publication bias was detected (Egger's test: $z = –0.05$, $p = 0.960$). Funnel plot in figure S21. Influence diagnostics (figure S22) indicated one potential outlier (Hamzeloo et al., 2016), but after removing the study the heterogeneity did not improve significantly $t^2 = 0.0497$.

*Moderators:* Combined model of variables and age did not significantly associate with effect size. Scales significantly moderate the prevalence ($F_{(5,6)} = 19.66$, $p = 0.001$). MCMI-II, MCMI-III, SAP, and SCID-II scales yielded significantly higher effect estimates, while the Clinical assessment (only 1 study) did not show a significant association. Heterogeneity was improved but still hight ($I^2 = 94.12$ %). Similarly setting was a significant moderator ($F_{(4,7)} = 18.15$, $p = 0.0008$). Higher effect estimates were observed for prison, ADHD service, and general adult outpatient samples, while the student sample showed no significant association. Heterogeneity was less but significant ($Q = 120.82$, df:7, $p < 0.001$, $I^2 = 93.89$ %).

### 3.2.9. Estimation of avoidant personality disorder prevalence

The pooled prevalence of Avoidant PD ($k = 10$, $n = 1973$, $e = 409$) was 23.1 % (95 % CI [14.7- 32.8]), with high heterogeneity ($I^2 = 91.6$ %, $Q = 106.63$, df:9, $p < 0.001$). Figure S23. Funnel plot (figure S24) and Egger's test: ($z = – 0.035$, $p = 0.970$) showed no publication bias. Influence diagnostics (figure S5) suggested one study as potentially influential, (Irastorza Eguskiza et al. (2018) and when it excluded, this significantly reduced heterogeneity ($\tau^2 = 0.0107$).

*Moderators:* Combined Model and age did not show significant effects. Scales significantly moderate the prevalence ($F_{(5,5)} = 13.81$, $p = 0.006$), but heterogeneity ($I^2 = 96.5$ %) was high. Setting significantly moderated prevalence estimates ($F_{(3,7)} = 29.01$, $p = 0.0003$), with high residual heterogeneity ($I^2 = 93.5$ %). Prevalence was significantly higher in ADHD services and general adult outpatient samples ($p < 0.001$), compared to students.

### 3.2.10. Estimation of Dependent Personality Disorder prevalence

The random-effects model ($k = 10$, $n = 1785$; $e = 204$) estimated a pooled prevalence of 15.2 % (95 % CI: 7.8–24.4). Figure S25. Heterogeneity was substantial ($I^2 = 95.8$ %, $Q = 213.33$, df:9, $p < 0.0001$). No evidence of publication bias was detected (Egger's test: $z = 0.41$, $p = 0.680$), and funnel plot in figure S26. Influence diagnostics (figure S27) did not identify any study as influential.

*Moderators*: Age and combined variables shown no significant effect. Scales had a significant moderator effect ($F_{(5,5)} = 9.31$, $p = 0.014$) but high heterogeneity ($I^2 = 95.3$ %). MCMI-II, MCMI-III, and SCID-II yielded significantly higher effect estimates compared to others (Clinical, SAP). Similarly, the setting had a significant moderating effect ($F_{(4,6)} = 8.85$, $p = 0.0109$). ADHD services setting ($p = 0.0028$) and general adult outpatient setting ($p = 0.0365$) showed significantly higher effect sizes, while prison and student samples did not differ significantly.

### 3.2.11. Estimation of obsessive-compulsive personality disorder prevalence

The random-effects meta-analysis ($k = 10$, $n = 1973$, $e = 324$) estimated a pooled prevalence of 12.5 % (95 % CI: 5.1–22.5), with substantial heterogeneity ($I^2 = 98.2$ %, $p < 0.0001$). Figure S29. No evidence of publication bias was detected (Egger's test $z = –0.50$, $p = 0.620$), funnel plot (figure S30). Influence diagnostics (figure S5) suggested one study as potentially influential, (Irastorza Eguskiza et al. (2018) and when it excluded, this significantly reduced heterogeneity ($\tau^2 = 0.021$).

*Moderators:* Combined Model and age were not significantly associate with prevalence. Scales also showed overall no significant association, but SCID-II showed a significant positive association ($\beta = 0.50$, $p = 0.0125$), suggesting it may be more sensitive to detect cases compared to other scales. Heterogeneity was high ($I^2 = 98.5$ %, $p < 0.0001$). Setting showed a significant moderator effect ($F_{(3,7)} = 10.97$, $p = 0.005$). Both the general adult outpatient ($\beta = 0.75$, $p = 0.0023$) and ADHD services ($\beta = 0.26$, $p = 0.0204$) settings had significant positive effects on the prevalence (higher). Residual heterogeneity was high ($I^2 = 96.35$ %, $p < 0.0001$).

Consistent with the reporting in the primary literature, this meta-analysis includes also some personality categories (e.g., Passive-Aggressive, Depressive, Masochistic, and Sadistic PD) that were featured in earlier diagnostic frameworks or specific psychometric tools (e.g., MCMI-II/III). While these are no longer recognized as standalone categorical diagnoses in the DSM-5, they are presented here to provide a comprehensive and transparent synthesis of the data reported in the original studies. These categories should be interpreted as exploratory and distinct from the current DSM-5 personality disorder clusters

### 3.2.12. Estimation of passive-aggressive personality disorder prevalence

The pooled prevalence ($k = 9$; $n = 1922$, $e = 374$) was 25.3 % (95 % CI [14.2–38.4]). Heterogeneity was substantial ($I^2 = 98.1$ %, $p < 0.0001$). Inspection of funeral plot (figure S33) and Egger's test ($z = 0.74$, $p = 0.460$) showed no significant evidence of publication bias. Influence diagnostics (figure S34) did not identify any study as influential.

*Moderators:* Combined model and age showed again no significant effect. Scales had a significant effect ($F_{(4,5)} = 9.25$, $p = 0.016$), with substantial heterogeneity ($I^2 = 98.1$ %). MCMI-II, MCMI-III, and SCID-II scales showed significant positive associations with the prevalence ($p < 0.05$), suggesting detecting higher rates compared to clinical. Setting was significant moderator ($F_{(4,5)} = 8.0997$, $p = 0.021$ with substantial heterogeneity ($I^2 = 97.87$ %). Higher rates of Passive-Aggressive PD observed in ADHD services and outpatient clinics of general adult psychiatry compared to prison and students' settings.

### 3.2.13. Estimation of depressive personality disorder prevalence

Depressive PD is not listed as a diagnosis in the DSM-5 but in previous versions (DSM-II and DSM-IV). Six studies report data for this PD ($k = 6$, $n = 1368$, $e = 242$). Pooled prevalence was 22.5 % (95 % CI: 9.3–39.2), see also figure S35, with extremely high heterogeneity ($I^2 = 97.9$ %, $Q = 153.40$, df:5, $p < 0.0001$). No evidence of publication bias was detected (Egger's test: $z = –0.049$, $p = 0.96$), funnel plot figure S36. Influence diagnostics (figure S37) indicate that the studies of Irastorza Eguskiza et al. (2018) and Kavakçi et al. (2012) exert notable influence on the overall model.

*Moderators:* Combined model and age shown no significant effects while scales had a significant moderator effect ($p = 0.042$). MCMI-III is associated with a higher estimated prevalence of depressive PD compared to the SCID-II. Heterogeneity was significant ($I^2 = 97.7$ %, $p < 0.0001$). Setting did not have any significant effect. ($F_{(4,2)} = 4.881$, $p = 0.177$). However, the heterogeneity remained high ($I^2 = 97.87$ %).

### 3.2.14. Estimation of other personality disorders prevalence

Some studies using the MCMI-III scale reported few more personality disorders which they are not listed in DSM-5 but in older versions. Masochistic PD is listed in the appendix of the DSM-III-R under the name Self-Defeating Personality Disorder but never in DSM editions. Is a category in the Milon's theory of personality disorders. Similarly Sadistic PD was included in the appendix of the DSM-III-R but was not included in later editions due to concerns about its overlap with other disorders like Antisocial PD and Narcissistic PD. Here, we will only report the pooled prevalence estimates and not analysis of moderator effects given the small number of studies.

Masochistic PD: The random-effects meta-analysis of four studies ($k = 4$, $n = 2120$, $e = 73$) found an overall prevalence of 14.3 % (95 % CI

[3.7- 30.0]), with substantial heterogeneity ($I^2 = 94.1$ %, $p < 0.001$). No evidence of publication bias was detected ($p = 0.840$).

Sadistic PD: ($k = 3$, $n = 321$, $e = 91$). The pooled prevalence was 28.9 % (95 % CI [6.1- 59.6]), with extremely high heterogeneity ($I^2 = 96.8$ %, $p < 0.001$). No evidence of publication bias was detected ($p = 0.690$).

For the five studies utilizing the SCID-II, diagnostic reliability was inferred based on the instrument's status as a gold-standard semi-structured interview; however, specific inter-rater reliability metrics (e.g., Cohen's Kappa) were not explicitly reported in the primary articles and thus could not be quantitatively synthesized.

### 3.3. Overview and synthesis of personality disorder prevalence

A summary of the meta-analytic results for all 15 personality disorder (PD) categories is presented in Table 2. Overall, prevalence estimates varied significantly across disorders, with Avoidant (23.1 %), Borderline (21.9 %), and Passive-Aggressive PD (25.3 %) emerging as the most frequent comorbidities. A consistent synthetic trend emerged across the models: statistical heterogeneity was high ($I^2$>90 %), driven primarily by the diagnostic instrument and the clinical setting. Specifically, the use of the SCID-II and assessments conducted within specialized ADHD services or general psychiatric outpatient clinics yielded significantly higher prevalence estimates compared to prison or student populations. While specific influence diagnostics and moderator analyses for each disorder are detailed in the following sections, the collective data in Table 2 highlights a substantial burden of Cluster B and C pathology within adult ADHD cohorts.

While Egger's test and funnel plots did not indicate significant publication bias, these results must be interpreted with extreme caution. Given the relatively small number of studies (k) in several analyses, these tests possess low statistical power and cannot definitively rule out the presence of publication bias.

## 4. Discussion

The results above indicate that the prevalence of co-existing PDs among adults with ADHD is considerable across multiple PD categories. When all PDs were considered, 57 % of adults with ADHD met criteria for at least one PD which far exceeds that observed in the general population, of around 7.8 % (95 % CI: 6.1–9.5) for any PD worldwide. (Winsper et al., 2019). Further, in Western-country general adult populations the estimate of any PD is about 12.16 % (Volkert et al., 2018) which again is much lesser than the PD rate in ADHD population. This substantial difference suggests that PD comorbidity is not only common but may represent a characteristic feature of adult ADHD presentations. However, it is important to note that the pooled prevalence for 'Any Personality Disorder' (57 %) was based on a subset of five studies that provided aggregate diagnostic data. These studies predominantly sampled from specialized adult ADHD outpatient services, potentially representing a higher degree of clinical complexity than the broader study pool. Consequently, this figure should be interpreted as an estimate of morbidity within high-severity clinical cohorts rather than a universal baseline for all adults with ADHD. This indicates the need for future primary studies to consistently report aggregate 'Any PD' rates alongside specific categories to allow for more robust meta-analytic synthesis

Specific PDs such as Avoidant (23.1 %), Passive-Aggressive/Negativistic (25.3 %), Borderline (22 %), and Depressive PD (22 %) also showed nontrivial prevalence rates. These findings are striking when contrasted with general population estimates—Avoidant PD:1.5 % and 2.5 % (Moini et al., 2024), Passive-Aggressive about 0.52 % (Millon, 1993; Widiger, 1991), Borderline PD 1.6–5.9 % (Grant et al., 2008), and Depressive PD around 3–6 % (Bagby et al., 2012; Klein and Miller, 1993; Klein, 1999; Ørstavik et al., 2007). The magnitude of these differences—often a five- to tenfold increase—suggests that adults with ADHD experience a markedly greater burden of affective and interpersonal dysregulation, consistent with clinical observations of chronic instability, emotional impulsivity, and rejection sensitivity in this population (Ditrich et al., 2021; Weiner et al., 2019).

Avoidant PD, defined by fear of rejection, social tension, chronic anxiety, feelings of inadequacy, and hypersensitivity to criticism, overlaps strongly with ADHD symptomatology (Faraone et al., 2015; Lampe and Malhi, 2018). Similarly, Depressive PD—characterized by low self-esteem, hopelessness, guilt, pessimism, and somatic rationalizations of incapacity (Davis and Patterson, 2005; Huprich, 1998)—shares core affective disturbances with ADHD, particularly emotional dysregulation, and chronic demoralization (Barkley and Fischer, 2010; Shaw et al., 2014). These shared traits suggest that some dependent features in ADHD may reflect maladaptive responses to chronic functional difficulties rather than distinct comorbid pathology.

**Table 2**
Summary of meta-analytic results for personality disorders in adults with ADHD.

| Personality Disorder | k | n | e | Pooled preval. % | 95 % CI | Significant Moderators | $I^2$ (%) |
|---|---|---|---|---|---|---|---|
| Any PD | 5 | 835 | 461 | 57 | 42.0–71.0 | Scale (MCMI-III > SCID-II > Clinical, $p = 0.017$) | 98 |
| Schizotypal PD | 8 | 1608 | 83 | 7.7 | 1.1–18.8 | None | 97.3 |
| Paranoid PD | 10 | 1785 | 220 | 11.9 | 5.6–20.1 | Scale (SCID-II, $p = 0.016$); setting (ADHD/General > Prison/Students, $p = 0.0025$) | 98.2 |
| Schizoid PD | 9 | 1638 | 85 | 6.4 | 1.4–14.1 | Scale (MCMI-III, $p = 0.043$); setting (ADHD > Students, $p = 0.042$) | 98.1 |
| Borderline PD | 11 | 2120 | 421 | 21.9 | 12.4–33.1 | Scale (SCID-II, $p = 0.011$); setting (ADHD/Outpatient > Prison/Students, $p = 0.006$) | 97.7 |
| Antisocial PD | 11 | 2120 | 289 | 18.2 | 10.3–27.5 | Scale (MCMI-II/III, SAP, SCID-II, $p < 0.01$); setting (Prison > ADHD > Outpatient, $p < 0.01$) | 98.2 |
| Histrionic PD | 9 | 1638 | 271 | 7.3 | 1.0–17.9 | Scale (MCMI-II, SCID-II, $p < 0.05$) | 97.4 |
| Narcissistic PD | 9 | 1943 | 368 | 13.5 | 2.5–30.7 | Scale (MCMI-II/III, SAP, SCID-II, $p = 0.0012$); setting (Prison/ADHD > Students, $p < 0.001$) | 98.0 |
| Avoidant PD | 10 | 1973 | 409 | 23.1 | 14.7–32.8 | Scale (MCMI-II/III, SCID-II, $p = 0.006$); setting (ADHD/Outpatient > Students, $p = 0.0003$) | 96.5 |
| Dependent PD | 10 | 1785 | 204 | 15.2 | 7.8–24.4 | Scale (MCMI-II/III, SCID-II, $p = 0.014$); setting (ADHD/Outpatient > Others, $p = 0.011$) | 96.3 |
| Obsessive–Compulsive PD | 10 | 1973 | 324 | 12.5 | 5.1–22.5 | Scale (SCID-II, $p = 0.0125$); setting (ADHD/Outpatient > Students, $p = 0.005$) | 98.2 |
| Passive–Aggressive PD | 9 | 1922 | 374 | 25.3 | 14.2–38.4 | Scale (MCMI-II/III, SCID-II, $p = 0.016$); setting (ADHD/Outpatient > Others, $p = 0.021$) | 98.1 |
| Depressive PD | 6 | 1368 | 242 | 22.5 | 9.3–39.2 | Scale (MCMI-III > SCID-II, $p = 0.042$) | 97.9 |
| Masochistic PD | 4 | 2120 | 73 | 14.3 | 3.7–30.0 | — Limited data | 94.1 |
| Sadistic PD | 3 | 321 | 91 | 28.9 | 6.1–59.6 | — Limited data | 96.8 |

**Note.** $k$ = number of studies; $n$ = total sample size; $e$ = number of events; CI = confidence interval. All models used random-effects estimators. All $I^2$ values > 90 %, indicating substantial heterogeneity. Egger's tests for publication bias were non-significant across analyses.

Borderline PD and ADHD also exhibit substantial overlap in impulsivity, emotional dysregulation, and unstable relationships (Matthies and Philipsen, 2014; Philipsen et al., 2008). Both involve rapid mood shifts, low frustration tolerance, and impaired inhibition linked to fronto-limbic dysfunction and neurotransmitter dysregulation (Skirrow, 2012; Asherson et al., 2014; Kebets et al., 2021; Reimherr et al., 2005; Weiner et al., 2019). BPD-like patterns may develop secondarily to chronic ADHD-related failures, rejection sensitivity, and unstable self-concept (Ferrer et al., 2010), supporting the hypothesis that early ADHD symptoms can predispose to later borderline pathology (van Dijk and Anckarsäter, 2011).

In addition, Antisocial and Passive-Aggressive PDs, marked by impulsivity, anger, and defiance (Millon et al., 2009), also co-occur frequently with ADHD. Longitudinal data link childhood conduct and oppositional defiant disorders to adult antisocial PD in ADHD populations (Biederman et al., 2008; Dowson, 2008; Rosa-Justicia et al., 2022).

Furthermore, Dependent PD, observed in about 15 % of adults with ADHD versus 0.5–0.6 % in the general population (APA, 2013), involves submissive, clinging behaviour and fear of separation. These interpersonal difficulties parallel ADHD-related dependence on external structure and reassurance (Skirrow et al., 2009). Chronic executive dysfunction and low self-efficacy may reinforce dependency as a compensatory adaptation (Barkley and Fischer, 2010; Shaw et al., 2014). Such parallels suggest that dependent traits in ADHD populations may emerge as adaptive or compensatory responses to persistent executive dysfunction and emotional instability rather than as distinct personality pathology.

Obsessive–Compulsive PD (OCPD) prevalence was also elevated. Though seemingly opposite to ADHD, perfectionism and rigidity may emerge as compensatory mechanisms for chronic disorganization (Barra et al., 2021; Josephson et al., 2007; Young, 2005). Shared deficits in cognitive flexibility and executive control (Anholt, 2010; Abramovitch et al., 2013; Barkley, 2015) suggest OCPD-like traits can arise as maladaptive strategies to manage ADHD-related chaos (Anholt, 2014; Barra et al., 2021).

Finally, Schizotypal, Paranoid, and Schizoid PDs were more prevalent among ADHD populations than community estimates of 3.9–4.6 % (Lenzenweger, 2008; Torgersen et al., 2001). These elevations may reflect overlapping neurodevelopmental vulnerabilities, including attentional bias, social-cognitive impairment, and executive dysfunction (Skirrow, 2012).

Overall, the markedly elevated PD prevalence highlights the complex interplay between neurodevelopmental deficits, emotional regulation difficulties, and maladaptive personality structures.

Moderator analyses indicated that both measurement scales and study settings significantly influenced prevalence estimates. Studies employing structured diagnostic interviews such as the SCID-II or self-report inventories like the MCMI-III reported higher PD prevalence, likely reflecting greater sensitivity to subclinical or dimensional traits (Millon and Davis 1997). Similarly, prevalence was higher in ADHD specialty and general psychiatric clinics than in student or prison samples, suggesting that clinical settings capture more impaired populations, though even non-clinical samples showed elevated PD rates. Collectively, these findings support a robust multidimensional association between ADHD and PDs—particularly those characterized by affective dysregulation and interpersonal sensitivity—while highlighting how diagnostic approach and context shape observed heterogeneity.

### 4.1. Interpretation of prevalence & heterogeneity

The elevated rate of “any PD” in ADHD supports earlier narrative and systematic reviews noting a strong association between ADHD and PDs. For example, Matthies and Philipsen (2016) reported ADHD as a major risk factor for adult PDs, while prevalence of borderline PD among adults with ADHD has exceeded 30 % in other studies (Matthies and Philipsen, 2014). However, these results must be interpreted in light of the extremely high heterogeneity ($I^2$ mostly $> 90$ %) observed across PD categories. The large between-study variability perhaps reflects differences in diagnostic instruments, sample settings, and how ADHD and PDs were operationalised. The meta-regression indicated that diagnostic scale (e.g., MCMI-III vs SCID-II vs clinical assessment) significantly moderated prevalence estimates in several PDs, suggesting that the choice of diagnostic instrument significantly influenced prevalence estimates, with self-report measures like the MCMI-III consistently yielding higher rates than semi-structured interviews such as the SCID-II. This discrepancy is likely attributable to the inherent nature of self-report data, which may capture transient emotional states or symptoms of ADHD (e.g., impulsivity and affective lability) as PD, leading to potential over-identification. In contrast, semi-structured interviews allow clinicians to differentiate between episodic symptoms and the stable, long-term patterns of dysfunction required for a personality disorder diagnosis. Consequently, while self-report scales are valuable for screening, the estimates derived from interview-based studies in this meta-analysis likely provide a more accurate reflection of true clinical co-existence of ADHD and PDs.

### 4.2. The nature of prevalence variability

The dominant statistical finding of this meta-analysis is the extreme heterogeneity ($I^2>90$ %) across all personality disorder (PD) categories, which indicates that the "true" prevalence of PD in ADHD is highly context-dependent. Consequently, the primary contribution of this work is not the establishment of a single, precise prevalence figure, but rather the demonstration of how significantly these estimates fluctuate based on the categorical diagnostic instrument and clinical setting. These findings underscore the blurred boundaries between core ADHD symptomatology (e.g., impulsivity, emotional dysregulation) and categorical personality pathology.

The meta-regression indicated that the choice of diagnostic scale significantly moderated prevalence estimates; self-report categorical measures like the MCMI-III consistently yielded higher rates than semi-structured interviews such as the SCID-II. This discrepancy is likely attributable to the inherent nature of self-report data, which may capture transient emotional states or functional impairments of ADHD as PD-level symptoms, leading to potential over-identification. In contrast, semi-structured interviews allow clinicians to better differentiate between episodic neurodevelopmental symptoms and the stable, long-term patterns of dysfunction required for a formal, categorical PD diagnosis.

Therefore, when categorical tools are applied to ADHD populations, the high prevalence rates may reflect a methodological sensitivity to shared clinical distress rather than the presence of distinct, stable comorbid disorders. While self-report scales remain valuable for clinical screening, the estimates derived from interview-based studies in this meta-analysis likely provide a more accurate reflection of the true categorical co-existence of ADHD and personality pathology.

### 4.3. Theoretical implications

The high prevalence rates observed in this meta-analysis raise a critical conceptual question: do these figures represent distinct comorbid disorders or a phenotypic overlap inherent to current diagnostic criteria? Several mechanisms may explain the frequent co-occurrence of ADHD and PDs. First, as a neurodevelopmental disorder characterized by early-onset inattention, impulsivity, and executive dysfunction, ADHD may hinder adaptive personality development and contribute to the stabilization of maladaptive patterns. Specifically, ADHD-related impairments in the acquisition of stable coping and self-regulation skills may increase vulnerability to personality pathology, leading to the crystallization of PD-level dysfunction by adulthood (Matthies and Philipsen, 2016).

Second, overlapping symptoms—particularly emotional

dysregulation, impulsivity, and interpersonal instability—are shared by ADHD and Cluster B/C personality disorders. Adults with ADHD often exhibit heightened emotional reactivity and poor frustration tolerance, which can evolve into or be misidentified as personality-disordered patterns (Ditrich et al., 2021). This suggests that the high rates—particularly in Cluster B and C—may also reflect a diagnostic artifact, where symptoms of chronic ADHD-related distress and neurodevelopmental impairment are captured as personality pathology by standard categorical instruments.

Third, shared genetic and neurobiological factors may underlie both conditions. Although direct GWAS evidence remains limited, twin and phenotypic data indicate a substantial correlation, particularly between ADHD symptoms and borderline PD traits (($r \approx 0.59$; Matthies and Philipsen, 2014). Ultimately, while the underlying causal boundaries between neurodevelopmental symptoms and personality stability remain a subject for longitudinal investigation, this meta-analysis highlights the clinical reality that a majority of these individuals meet formalized diagnostic thresholds for both conditions when standard categorical tools are applied.

### 4.3. Clinical implications

The high rates of PDs in adults with ADHD emphasise the need for comprehensive assessment and integrated care. Clinicians treating adult ADHD should routinely screen for PDs, because the presence of a PD may complicate treatment planning (e.g., pharmacotherapy, adherence, psychotherapy). Structured instruments such as the SCID-II appeared more sensitive in detecting PDs in ADHD samples, according to our moderator analyses.

Moreover, the presence of a PD may impact treatment outcomes for ADHD: personality pathology may affect response to stimulant medication, increase risk of comorbidities (e.g., substance use, suicidality), and require adjunctive psychotherapeutic interventions. This highlights the importance of multidisciplinary approaches.

### 4.4. Methodological considerations

A critical methodological consideration in the study of ADHD and PD co-existence is the distinction between symptoms on a continuous spectrum and categorical clinical diagnoses. In this meta-analysis, the majority of included studies utilized formal clinical evaluations and validated scales that require meeting specific thresholds for symptom frequency, age of onset, and functional impairment. This rigor is essential to ensure that the observed comorbidities are not merely a reflection of overlapping traits—such as impulsivity or emotional lability—but represent the co-occurrence of distinct diagnostic entities. Future research should prioritize the use of longitudinal data and the LEAD procedure to further refine the differentiation of these complex clinical profiles. In addition, regarding diagnostic administration, the clinical context of the included research provides a proxy for expertise; all but two studies were situated in specialized ADHD or psychiatric outpatient services. Assessments in these settings are traditionally performed by clinicians or trained psychiatric professionals, significantly reducing the risk of prevalence inflation often associated with lay-interviewer administration. This clinical oversight supports the validity of the categorical diagnoses reported across the majority of the sampled cohorts. While clinical evaluation can introduce variability, all included studies utilized standardized diagnostic scales as a foundational component of the assessment process to ensure objective symptom thresholds were met. In the case of Gift et al. (2016), clinical evaluation was used to supplement scale data, a method that mirrors the LEAD (Longitudinal, Expert, All Data) gold-standard approach by combining psychometric structure with professional clinical judgment. This dual-method approach reduces the risk of misclassification compared to unstructured clinical assessments alone. Furthermore, a methodological problem is the lack of standardized reporting regarding key clinical moderators such as history of trauma, current treatment status (e.g., stimulant medication), and psychiatric comorbidities. While these factors significantly influence the expression and severity of personality pathology, they were not reported with sufficient consistency across the identified studies to permit a robust moderator analysis. Consequently, the role of these variables in the observed prevalence rates remains an important area for future primary investigation.

Finally, a potential methodological problem regarding diagnostic coherence is the inclusion of outdated personality categories such as Passive-Aggressive and Depressive PD. These were included to maintain fidelity to the primary data sources, as they constituted a significant portion of the pathology reported in earlier clinical cohorts. Their high prevalence in this meta-analysis likely reflects the specific sensitivity of Millon's evolutionary model to the chronic interpersonal and self-regulatory difficulties common in adult ADHD, though they should be viewed as exploratory constructs rather than current DSM-5 comorbidities.

### 4.5. Strengths and limitations

Key strengths of this meta-analysis include its comprehensive coverage of multiple PD categories in the ADHD adult population, systematic extraction of moderators, and use of rigorous meta-analytic techniques (e.g., inverse-double arcsine transformation, Knapp–Hartung adjustment for meta-regression). Nevertheless, there are important limitations. First, the high heterogeneity ($I^2 > 90$ %) reduces confidence in pooled point estimates and suggests that interpretation should focus more on broad trends rather than precise prevalences. Second, despite efforts to identify moderators, residual heterogeneity remained large in most analyses, indicating that unmeasured factors (e.g., trauma history, ADHD subtype, comorbidities, cultural differences) maybe also contribute. Third, many included studies were clinical rather than representative community samples, potentially inflating prevalence estimates. Fourth, temporal sequence (whether ADHD leads to PDs or co-develops) cannot be established from cross-sectional designs. Finally, a limitation of this review is the inclusion of studies utilizing validated diagnostic scales alongside those using gold-standard diagnostic interviews (LEAD procedure). While the former may include individuals with sub-threshold symptoms, this approach was necessary to synthesize the broader body of available research. To mitigate this, we have done the moderator analysis where it shows how each scale influences the prevalence.

## 5. Conclusion

In summary, adults with ADHD exhibit markedly elevated rates of co-occurring personality disorders across a broad spectrum. Although the extreme heterogeneity and the limited power of publication bias tests limit the precision of single pooled prevalence estimates, the overall pattern is consistent: ADHD rarely occurs in isolation in adulthood. These findings indicate that prevalence is highly context-dependent, fluctuating significantly based on the diagnostic instrument and clinical setting used. Clinicians should adopt diagnostic and therapeutic strategies that address both ADHD and concurrent personality pathology. Future research should clarify developmental pathways, shared mechanisms, and optimal interventions to improve long-term outcomes for this complex and high-risk population.

## Funding statement

This research received no specific grant from any funding agency, commercial or not-for-profit sectors

## Ethical approval and informed consent statements

N/A.

## CRediT authorship contribution statement

**Dimitrios Adamis:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Tianran Zhang:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation. **Blánaid Gavin:** Writing – review & editing, Writing – original draft, Supervision, Methodology. **Fiona McNicholas:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgements

Not applicable (N/A).

## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.psychres.2026.117085.

## Data availability

Data are available upon reasonable request from the corresponding author.

## References

Abramovitch, A., Dar, R., Mittelman, A., Schweiger, A., 2013. Don't judge a book by its cover: aDHD-like symptoms in obsessive compulsive disorder. J. Obsessive Compuls. Relat. Disord. 2 (1), 53–61.

Adamis, D., Kasianenko, D., Usman, M., Saleem, F., Wrigley, M., Gavin, B., McNicholas, F., 2023. Prevalence of personality disorders in adults with Attention Deficit Hyperactivity Disorder (ADHD). J. Atten. Disord. 27 (7), 658–668. https://doi.org/10.1177/10870547231161531.

American Psychiatric Association. 2013. Diagnostic and statistical manual of mental disorders (5th ed.).

Anholt, G., Aderka, I., Van Balkom, A., Smit, J., Schruers, K., Van Der Wee, N., Eikelenboom, M., De Luca, V., Van Oppen, P., 2014. Age of onset in obsessive–compulsive disorder: admixture analysis with a large sample. Psychol. Med. 44 (1), 185–194.

Anholt, G.E., Cath, D.C., van Oppen, P., Eikelenboom, M., Smit, J.H., van Megen, H., van Balkom, A.J., 2010. Autism and ADHD symptoms in patients with OCD: are they associated with specific OC symptom dimensions or OC symptom severity? J. Autism. Dev. Disord. 40 (5), 580–589. https://doi.org/10.1007/s10803-009-0922-1.

Asherson, P., Young, A.H., Eich-Höchli, D., Moran, P., Porsdal, V., Deberdt, W., 2014. Differential diagnosis, comorbidity, and treatment of attention-deficit/hyperactivity disorder in relation to bipolar disorder or borderline personality disorder in adults. Curr. Med. Res. Opin. 30 (8), 1657–1672. https://doi.org/10.1185/03007995.2014.915800.

Bagby, R.M., Watson, C., Ryder, A.G., 2012. Depressive personality disorder. In: Widiger, T.A. (Ed.), The Oxford Handbook of Personality Disorders (pp. 0). Oxford University Press. https://doi.org/10.1093/oxfordhb/9780199735013.013.0029.

Balduzzi, S., Rücker, G., Schwarzer, G., 2019. How to perform a meta-analysis with R: a practical tutorial. Evid. Based. Ment. Health 22 (4), 153–160.

Barkley, R.A., 2015. Attention-deficit Hyperactivity disorder: A handbook For Diagnosis and treatment, 4th Ed (chapter 5). The Guilford Press.

Barkley, R.A., Fischer, M., 2010. The unique contribution of emotional impulsiveness to impairment in major life activities in hyperactive children as adults. J. Am. Acad. Child Adolesc. Psychiatry 49 (5), 503–513. https://doi.org/10.1097/00004583-201005000-00011.

Barra, S., Grub, A., Roesler, M., Retz-Junginger, P., Philipp, F., Retz, W., 2021. The role of stress coping strategies for life impairments in ADHD. J. Neural. Transm. 128 (7), 981–992.

Biederman, J., DiSalvo, M., Woodworth, K.Y., Fried, R., Uchida, M., Biederman, I., Spencer, T.J., Surman, C., Faraone, S.V., 2020. Toward operationalizing deficient emotional self-regulation in newly referred adults with ADHD: a receiver operator characteristic curve analysis. Eur. Psychiatry 63 (1), e21. https://doi.org/10.1192/j.eurpsy.2019.11.

Biederman, J., Petty, C.R., Dolan, C., Hughes, S., Mick, E., Monuteaux, M.C., Faraone, S.V., 2008. The long-term longitudinal course of oppositional defiant disorder and conduct disorder in ADHD boys: findings from a controlled 10-year prospective longitudinal follow-up study. Psychol. Med. 38 (7), 1027–1036. https://doi.org/10.1017/S0033291707002668.

Cherkasova, M.V., Roy, A., Molina, B.S.G., Scott, G., Weiss, G., Barkley, R.A., Biederman, J., Uchida, M., Hinshaw, S.P., Owens, E.B., Hechtman, L., 2022. Review: adult outcome as seen through controlled prospective follow-up studies of children with Attention-Deficit/Hyperactivity disorder followed into adulthood. J. Am. Acad. Child Adolesc. Psychiatry 61 (3), 378–391. https://doi.org/10.1016/j.jaac.2021.05.019.

Cumyn, L., French, L., Hechtman, L., Cumyn, L., French, L., Hechtman, L., 2009. Comorbidity in adults with attention-deficit hyperactivity disorder. Can. J. Psychiatry 54 (10), 673–683. https://doi.org/10.1177/070674370905401004.

Davis, R.D., Patterson, M.J., 2005. Diagnosing personality disorder subtypes with the MCMI-III. In: Craig, R.J. (Ed.), New Directions in Interpreting the Millon Clinical Multiaxial Inventory–III (MCMI-III). John Wiley & Sons, pp. 32–70.

Ditrich, I., Philipsen, A., Matthies, S., 2021. Borderline personality disorder (BPD) and attention deficit hyperactivity disorder (ADHD) revisited - a review-update on common grounds and subtle distinctions. Borderline Personal. Disord. Emot. Dysregul. 8 (1), 22. https://doi.org/10.1186/s40479-021-00162-w.

Dowson, J.H., 2008. Characteristics of adults with attention-deficit/hyperactivity disorder and past conduct disorder. Acta Psychiatr. Scand. 117 (4), 299–305. https://doi.org/10.1111/j.1600-0447.2008.01153.x.

Edvinsson, D., Lindström, E., Bingefors, K., Lewander, T., Ekselius, L., 2013. Gender differences of axis I and II comorbidity in subjects diagnosed with attention-deficit hyperactivity disorder as adults [Article]. Acta Neuropsychiatr. 25 (3), 165–174. https://www.scopus.com/inward/record.uri?eid=2-s2.0-84879955388&partnerID=40&md5=b9078e014d107ea96234193fbe34cbc1.

Faraone, S.V., Asherson, P., Banaschewski, T., Biederman, J., Buitelaar, J.K., Ramos-Quiroga, J.A., Rohde, L.A., Sonuga-Barke, E.J.S., Tannock, R., Franke, B., 2015. Attention-deficit/hyperactivity disorder. Nat. Rev. Dis. Primers 1 (1), 15020. https://doi.org/10.1038/nrdp.2015.20.

Ferrer, M., Andión, O., Matalí, J., Valero, S., Navarro, J.A., Ramos-Quiroga, J.A., Torrubia, R., Casas, M., 2010. Comorbid attention-deficit/hyperactivity disorder in borderline patients defines an impulsive subtype of borderline personality disorder. J. Pers. Disord. 24 (6), 812–822. https://doi.org/10.1521/pedi.2010.24.6.812.

Gift, T.E., Reimherr, F.W., Marchant, B.K., Steans, T.A., Wender, P.H., 2016. Personality disorder in adult Attention-Deficit/Hyperactivity Disorder: attrition and change during long-term treatment. J. Nerv. Ment. Dis. 204 (5), 355–363. https://doi.org/10.1097/nmd.0000000000000470.

Gomez, R., Corr, P.J., 2014. ADHD and personality: a meta-analytic review. Clin. Psychol. Rev. 34 (5), 376–388. https://doi.org/10.1016/j.cpr.2014.05.002.

Grant, B.F., Chou, S.P., Goldstein, R.B., Huang, B., Stinson, F.S., Saha, T.D., Smith, S.M., Dawson, D.A., Pulay, A.J., Pickering, R.P., Ruan, W.J., 2008. Prevalence, correlates, disability, and comorbidity of DSM-IV borderline personality disorder: results from the Wave 2 national epidemiologic survey on alcohol and related conditions. J. Clin. Psychiatry 69 (4), 533–545. https://doi.org/10.4088/jcp.v69n0404.

Hamzeloo, M., Mashhadi, A., Salehi Fadardi, J., 2016. The prevalence of ADHD and comorbid disorders in Iranian adult male prison inmates. J. Atten. Disord. 20 (7), 590–598. https://doi.org/10.1177/1087054712457991.

Higgins, J., Thompson, S., Deeks, J., Altman, D., 2002. Statistical heterogeneity in systematic reviews of clinical trials: a critical appraisal of guidelines and practice. J. Health Serv. Res. Policy. 7 (1), 51–61. https://doi.org/10.1258/1355819021927674.

Hozo, S.P., Djulbegovic, B., Hozo, I., 2005. Estimating the mean and variance from the median, range, and the size of a sample. BMC Med. Res. Methodol. 5, 13. https://doi.org/10.1186/1471-2288-5-13.

Huprich, S.K., 1998. Depressive personality disorder: theoretical issues, clinical findings, and future research questions. Clin. Psychol. Rev. 18 (5), 477–500.

Irastorza Eguskiza, L.J., Bellón, J.M., Mora, M., 2018. Comorbidity of personality disorders and attention-deficit hyperactivity disorder in adults. Rev. Psiquiatr. Salud. Ment. (Engl. Ed) 11 (3), 151–155. https://doi.org/10.1016/j.rpsm.2016.01.009 (Comorbilidad de trastornos de personalidad y trastorno por déficit de atención e hiperactividad en adultos.).

Jacob, C.P., Groß-Lesch, S., Reichert, S., Geissler, J., Jans, T., Kittel-Schneider, S., Nguyen, T.T., Romanos, M., Reif, A., Dempfle, A., 2016. Sex- and subtype-related differences of personality disorders (Axis II) and personality traits in persistent ADHD [Article]. J. Atten. Disord. 20 (12), 1056–1065. https://www.scopus.com/inward/record.uri?eid=2-s2.0-84994213403&partnerID=40&md5=ae9ada6a5dead3ee8405db89c66f4007.

Jacobsson, P., Hopwood, C.J., Soderpalm, B., Nilsson, T., 2021. Adult ADHD and emerging models of maladaptive personality: a meta-analytic review. BMC Psychiatry 21 (1), 282. https://doi.org/10.1186/s12888-021-03284-1.

Josephson, S.C., Hollander, E., Sumner, J., 2007. Comprehensive treatment of three patients with comorbid OCPD and ADHD. CNS Spectr. 12 (5), 338–342. https://doi.org/10.1017/s1092852900021131.

Kavakçi, O., Kuğu, N., Semiz, M., Meydan, F., Karşikaya, S., Doğan, O., 2012. Prevalence of attention-deficit/hyperactivity disorder and co-morbid disorders among students of Cumhuriyet University [Article]. Eur. J. Psychiatry 26 (2), 107–117. https://www.scopus.com/inward/record.uri?eid=2-s2.0-84863803704&partnerID=40&md5=5c470836f706375bdacafa2015d64306.

Kebets, V., Favre, P., Houenou, J., Polosan, M., Perroud, N., Aubry, J.M., Van De Ville, D., Piguet, C., 2021. Fronto-limbic neural variability as a transdiagnostic correlate of emotion dysregulation. Transl. Psychiatry 11 (1), 545. https://doi.org/10.1038/s41398-021-01666-3.

Klein, D.N., 1999. Depressive personality in the relatives of outpatients with dysthymic disorder and episodic major depressive disorder and normal controls. J. Affect. Disord. 55 (1), 19–27. https://doi.org/10.1016/s0165-0327(98)00195-5.
Klein, D.N., Miller, G.A., 1993. Depressive personality in nonclinical subjects. Am. J. Psychiatry 150 (11), 1718–1724. https://doi.org/10.1176/ajp.150.11.1718.
Kooij, S.J., Bejerot, S., Blackwell, A., Caci, H., Casas-Brugue, M., Carpentier, P.J., Edvinsson, D., Fayyad, J., Foeken, K., Fitzgerald, M., Gaillac, V., Ginsberg, Y., Henry, C., Krause, J., Lensing, M.B., Manor, I., Niederhofer, H., Nunes-Filipe, C., Ohlmeier, M.D., Oswald, P., Pallanti, S., Pehlivanidis, A., Ramos-Quiroga, J.A., Rastam, M., Ryffel-Rawak, D., Stes, S., Asherson, P., 2010. European consensus statement on diagnosis and treatment of adult ADHD: the European Network Adult ADHD. BMC Psychiatry 10, 67. https://doi.org/10.1186/1471-244X-10-67.
Lampe, L., Malhi, G.S., 2018. Avoidant personality disorder: current insights. Psychol. Res. Behav. Manage 11, 55–66. https://doi.org/10.2147/prbm.S121073.
Lenzenweger, M.F., 2008. Epidemiology of personality disorders. Psychiatric Clin. North Am. 31 (3), 395–403. https://doi.org/10.1016/j.psc.2008.03.003.
Lobbestael, J., Leurgans, M., Arntz, A., 2011. Inter-rater reliability of the Structured Clinical interview for DSM-IV Axis I disorders (SCID I) and Axis II disorders (SCID II). Clin. Psychol. PsychOther 18 (1), 75–79. https://doi.org/10.1002/cpp.693.
Maffei, C., Fossati, A., Agostoni, I., Barraco, A., Bagnato, M., Deborah, D., Namia, C., Novella, L., Petrachi, M., 1997. Interrater reliability and internal consistency of the structured clinical interview for DSM-IV axis II personality disorders (SCID-II), version 2.0. J. Pers. Disord. 11 (3), 279–284. https://doi.org/10.1521/pedi.1997.11.3.279.
Matthies, S., Philipsen, A., 2016. Comorbidity of personality disorders and adult Attention Deficit Hyperactivity disorder (ADHD)–review of recent findings. Curr. Psychiatry Rep. 18 (4), 33. https://doi.org/10.1007/s11920-016-0675-4.
Matthies, S.D., Philipsen, A., 2014. Common ground in Attention Deficit Hyperactivity Disorder (ADHD) and Borderline Personality Disorder (BPD)-review of recent findings. Borderline Personal. Disord. Emot. Dysregul. 1, 3. https://doi.org/10.1186/2051-6673-1-3.
May, B., Bos, J., 2000. Personality characteristics of ADHD adults assessed with the Millon Clinical Multiaxial Inventory—II: evidence of four distinct subtypes. J. Pers. Assess. 75 (2), 237–248. https://doi.org/10.1207/S15327752JPA7502_5.
Millon, T., 1993. Negativistic (passive-aggressive) personality disorder. J. Pers. Disord. 7 (1), 78–85.
Millon, T., Davis, R.D., 1997. The MCMI-III: present and future directions. J. Pers. Assess. 68 (1), 69–85. https://doi.org/10.1207/s15327752jpa6801_6.
Millon, T., Millon, C., Davis, R.D., Grossman, S., 2009. Millon Clinical Multiaxial inventory-III (MCMI-III). Manual. Pearson/PsychCorp.
Moher, D., Liberati, A., Tetzlaff, J., Altman, D.G., Group, P., 2009. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. PLoS Med. 6 (7), e1000097. https://doi.org/10.1371/journal.pmed.1000097.
Moini, J., LoGalbo, A., Ahangari, R., 2024. Chapter 20 - personality disorders. In: Moini, J., LoGalbo, A., Ahangari, R. (Eds.), Foundations of the Mind, Brain, and Behavioral Relationships. Academic Press, pp. 333–352. https://doi.org/10.1016/B978-0-323-95975-9.00001-9.
Munn, Z., Moola, S., Riitano, D., Lisy, K., 2014. The development of a critical appraisal tool for use in systematic reviews addressing questions of prevalence. Int. J. Health Policy. Manage 3 (3), 123–128. https://doi.org/10.15171/ijhpm.2014.71.
Oliva, F., Mangiapane, C., Nibbio, G., Portigliatti Pomeri, A., Maina, G., 2020. MCMI-III personality disorders, traits, and profiles in Adult ADHD outpatients. J. Atten. Disord. 24 (6), 830–839. https://doi.org/10.1177/1087054718780319.
Ørstavik, R.E., Kendler, K.S., Czajkowski, N., Tambs, K., Reichborn-Kjennerud, T., 2007. The relationship between depressive personality disorder and major depressive disorder: a population-based twin study. Am. J. Psychiatry 164 (12), 1866–1872. https://doi.org/10.1176/appi.ajp.2007.07010045 quiz 1924.
Philipsen, A., Limberger, M.F., Lieb, K., Feige, B., Kleindienst, N., Ebner-Priemer, U., Barth, J., Schmahl, C., Bohus, M., 2008. Attention-deficit hyperactivity disorder as a potentially aggravating factor in borderline personality disorder. Br. J. Psychiatry 192 (2), 118–123. https://doi.org/10.1192/bjp.bp.107.035782.
Pilkonis, P.A., Heape, C.L., Ruddy, J., Serrao, P., 1991. Validity in the diagnosis of personality disorders: the use of the LEAD standard. Psychol. Assess.: A J. Consulting Clin. Psychol. 3 (1), 46.
Reimherr, F.W., Marchant, B.K., Strong, R.E., Hedges, D.W., Adler, L., Spencer, T.J., West, S.A., Soni, P., 2005. Emotional dysregulation in adult ADHD and response to atomoxetine. Biol. Psychiatry 58 (2), 125–131. https://doi.org/10.1016/j.biopsych.2005.04.040.
Rosa-Justicia, M., Saam, M.C., Flamarique, I., Borras, R., Naaijen, J., Dietrich, A., Hoekstra, P.J., Banaschewski, T., Aggensteiner, P., Craig, M.C., Sethi, A., Santosh, P., Sagar-Ouriaghli, I., Arango, C., Penzol, M.J., Brandeis, D., Werhahn, J.E., Glennon, J. C., Franke, B., Zwiers, M.P., Buitelaar, J.K., Schulze, U.M.E., Castro-Fornieles, J., 2022. Subgrouping children and adolescents with disruptive behaviors: symptom profiles and the role of callous-unemotional traits. Eur. Child Adolesc. Psychiatry 31 (1), 51–66. https://doi.org/10.1007/s00787-020-01662-w.
Shaw, P., Stringaris, A., Nigg, J., Leibenluft, E., 2014. Emotion dysregulation in attention deficit hyperactivity disorder. Am. J. Psychiatry 171 (3), 276–293. https://doi.org/10.1176/appi.ajp.2013.13070966.
Skirrow, C., McLoughlin, G., Kuntsi, J., Asherson, P., 2009. Behavioral, neurocognitive and treatment overlap between attention-deficit/hyperactivity disorder and mood instability. Expert. Rev. NeurOther 9 (4), 489–503. https://doi.org/10.1586/ern.09.2.
Skirrow, C., 2012. Investigating emotional lability in adults with Attention Deficit Hyperactivity Disorder: an integrative approach. PhD Thesis. University of London, King's College London, Institute of Psychiatry.
Skodol, A.E., Gunderson, J.G., Shea, M.T., McGlashan, T.H., Morey, L.C., Sanislow, C.A., Bender, D.S., Grilo, C.M., Zanarini, M.C., Yen, S., Pagano, M.E., Stout, R.L., 2005. The Collaborative Longitudinal Personality Disorders Study (CLPS): overview and implications. J. Pers. Disord. 19 (5), 487–504. https://doi.org/10.1521/pedi.2005.19.5.487.
Torgersen, S., Kringlen, E., Cramer, V., 2001. The prevalence of personality disorders in a community sample. Arch. Gen. Psychiatry 58 (6), 590–596. https://doi.org/10.1001/archpsyc.58.6.590.
Turgay, A., Goodman, D.W., Asherson, P., Lasser, R.A., Babcock, T.F., Pucci, M.L., Barkley, R., Group, A.T.P.M.W., 2012. Lifespan persistence of ADHD: the life transition model and its application. J. Clin. Psychiatry 73 (2), 192–201. https://doi.org/10.4088/JCP.10m06628.
van Dijk, F.E., Anckarsäter, H., 2011. ADHD, personality, and its disorders. In: Buitelaar, J.K., Kan, C.C., Asherson, P. (Eds.), ADHD in Adults: Characterization, Diagnosis, and Treatment. Cambridge University Press, pp. 174–190. https://doi.org/10.1017/CBO9780511780752.017.
Viechtbauer, W., 2010. Conducting meta-analyses in R with the metafor package. J. Stat. Softw. 36 (3), 1–48.
Volkert, J., Gablonski, T.C., Rabung, S., 2018. Prevalence of personality disorders in the general adult population in Western countries: systematic review and meta-analysis. Br. J. Psychiatry 213 (6), 709–715. https://doi.org/10.1192/bjp.2018.202.
Wan, X., Wang, W., Liu, J., Tong, T., 2014. Estimating the sample mean and standard deviation from the sample size, median, range and/or interquartile range. BMC Med. Res. Methodol. 14, 135. https://doi.org/10.1186/1471-2288-14-135.
Weiner, L., Perroud, N., Weibel, S., 2019. Attention deficit hyperactivity disorder and borderline personality disorder in adults: a review of their links and risks. Neuropsychiatr. Dis. Treatm. 3115–3129.
Widiger, T.A., 1991. Special section DSM-IV Reviews of the Personality Disorders: introduction to Special series. J. Pers. Disord. 5 (2), 122–134.
Winsper, C., Bilgin, A., Thompson, A., Marwaha, S., Chanen, A.M., Singh, S.P., Wang, A., Furtado, V., 2020. The prevalence of personality disorders in the community: a global systematic review and meta-analysis. Br. J. Psychiatry 216 (2), 69–78. https://doi.org/10.1192/bjp.2019.166.
Yadav, S.K., Bhat, A.A., Hashem, S., Nisar, S., Kamal, M., Syed, N., Temanni, M.R., Gupta, R.K., Kamran, S., Azeem, M.W., Srivastava, A.K., Bagga, P., Chawla, S., Reddy, R., Frenneaux, M.P., Fakhro, K., Haris, M., 2021. Genetic variations influence brain changes in patients with attention-deficit hyperactivity disorder. Transl. Psychiatry 11 (1), 349. https://doi.org/10.1038/s41398-021-01473-w.
Young, S., 2005. Coping strategies used by adults with ADHD. Pers. Individ. Dif. 38 (4), 809–816.
Young, S., Toone, B., Tyson, C., 2003. Comorbidity and psychosocial profile of adults with Attention Deficit Hyperactivity Disorder. Pers. Individ. Dif. 35 (4), 743–755. https://www.scopus.com/inward/record.uri?eid=2-s2.0-0041841547&partnerID=40&md5=9845733d34f3d589559d6dc5d9a57dd8.
Zimmerman, M., 1994. Diagnosing personality disorders. A review of issues and research methods. Arch. Gen. Psychiatry 51 (3), 225–245. https://doi.org/10.1001/archpsyc.1994.03950030061006.